



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-034A TransitCare Gate Review

Protected evidence diagnosis; no copy-ready solved project

Artifact	C3_034A_TRANSITCARE_REVIEW_PROTECTED_RC1
Status	PROTECTED - OPEN AFTER FROZEN ATTEMPT
Phase	C3-02 Transformation Engineering & Warehousing 14 lessons 34-46 focused hours
Prerequisite	C3-01 Gate PASS. C2B warehouse/modeling foundations may be assumed; implementation depth is taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: analogy -> tiny example -> guided real tool -> break/fix -> independent changed version -> validation
-> production reasoning -> evidence. A successful dbt command alone never proves correct data.



Review one competency at a time

Mark PASS only from saved evidence. If one required competency fails, Gate remains not passed. Diagnose the smallest failed contract; targeted repair does not automatically assign a fresh full retest.



T1 - Transformation & staging contracts

Layer boundaries, source/staging contracts and grains are explicit/replayable.

IF WEAK

Return only to the lesson(s) that teach this competency, rebuild one tiny example, close material, then complete a changed micro-retest. Save it as new remediation evidence; never edit the Gate attempt.

RETURN CONDITION

Changed evidence independently satisfies the competency and no linked critical failure remains.

T2 - Deterministic models & DAG

Dependencies and current-state selection are deterministic and reviewable.

IF WEAK

Return only to the lesson(s) that teach this competency, rebuild one tiny example, close material, then complete a changed micro-retest. Save it as new remediation evidence; never edit the Gate attempt.

RETURN CONDITION

Changed evidence independently satisfies the competency and no linked critical failure remains.

T3 - Testing & integrity

Grain/required/relationship/business tests block true defects; reconciliations prove no silent loss/fan-out.

IF WEAK

Return only to the lesson(s) that teach this competency, rebuild one tiny example, close material, then complete a changed micro-retest. Save it as new remediation evidence; never edit the Gate attempt.

RETURN CONDITION

Changed evidence independently satisfies the competency and no linked critical failure remains.

T4 - Incremental & recovery

Update/insert + unchanged rerun safe; rebuild/full-refresh recovery documented.

IF WEAK

Return only to the lesson(s) that teach this competency, rebuild one tiny example, close material, then complete a changed micro-retest. Save it as new remediation evidence; never edit the Gate attempt.

RETURN CONDITION

Changed evidence independently satisfies the competency and no linked critical failure remains.

T5 - History

Required descriptive change preserved as legitimate current+historical truth.

IF WEAK

Return only to the lesson(s) that teach this competency, rebuild one tiny example, close material, then complete a changed micro-retest. Save it as new remediation evidence; never edit the Gate attempt.

RETURN CONDITION

Changed evidence independently satisfies the competency and no linked critical failure remains.

T6 - Dimensional/late-reference integrity

Facts/dimensions preserve declared grain; late references handled without silent fact loss.

IF WEAK



Return only to the lesson(s) that teach this competency, rebuild one tiny example, close material, then complete a changed micro-retest. Save it as new remediation evidence; never edit the Gate attempt.

RETURN CONDITION

Changed evidence independently satisfies the competency and no linked critical failure remains.

T7 - Performance/CI/handoff

Plan reasoning, blocking CI/change safety and reproducible serving handoff are evidenced.

IF WEAK

Return only to the lesson(s) that teach this competency, rebuild one tiny example, close material, then complete a changed micro-retest. Save it as new remediation evidence; never edit the Gate attempt.

RETURN CONDITION

Changed evidence independently satisfies the competency and no linked critical failure remains.



Critical-failure audit

- Wrong fact grain, silent fan-out/loss or materially wrong accepted measure.
- Incremental logic duplicates business identities, misses required update, or same-delta rerun is unsafe.
- Required history overwritten or false versions fabricated.
- A true integrity test is disabled/weakened instead of repaired.
- Late-dimension fact disappears through uncontrolled join/filter.
- Schema/change behavior silently accepted without documented compatibility/rebuild implication.
- Performance change alters business semantics.
- Fabricated dbt/DuckDB/EXPLAIN/test evidence.
- Real secret/private data committed, logged or included in case/evidence package.
- Project cannot be reconstructed from documented source/dependency/setup state.

NEXT ROUTE

Gate A targeted weaknesses -> remediation + changed micro-retest. Gate B/C remain sealed unless an explicit full-retest assignment is issued. A strong average is irrelevant.